A New Efficient Alert Model for Disaster Management

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Abstract — During the various catastrophic events of recent years, The use of social media to communicate timely information during crisis situations has become a common practice in recent years, enabling the affected population to quickly publish a considerable amount of information about disasters, to help managers in making good and quick decisions.

We present a new real-time alert model for disaster management of a natural or anthropogenic disaster. This model is based on a neural network that integrates encapsulations from multiple sources, and thus retrieves information by combining multiple search results. Preliminary results are satisfactory.

Keywords — Disaster management, social networks, alert, knowledge

I. Introduction

In recent years, social networks have emerged as a potential resource for communicating, detecting, tracking and extracting disaster information to improve the management of both natural and anthropogenic disasters [1], [2], [3], [4], [5]. With the ability to share a message with a potentially important audience, Social networks such as Facebook, Twitter are used to collect and disseminate relevant and timely information [4], [5]. However, the amount of information now being communicated during these critical time and security situations can make it difficult to locate useful and usable information. Social networks are also used to sabotage stories such as the crash of the Asiana flight in July 2013, a photo of which was posted on Twitter 30 seconds after the crash [6], [7].

During the various catastrophic events of recent years - from forest fires in California in 2007 to the earthquake in Haiti in 2010, to the floods in Thailand in 2011, Typhoon Haiyan in the Philippines in 2013, the flood of the Elbe in June 2013 in Germany, Community Tweets - abusive messages targeting specific religious / racial groups exposed to natural disasters - social media enabled the affected population to quickly publish a considerable amount of information about disasters, to help managers make good and quick decisions [5], [19], [21], [22].

The use of social media to communicate timely information during crisis situations has become a common practice in recent years [3]. Major crises can generate millions of messages on social networks (*for example, 3.5 million messages in one day, during Hurricane Sandy in 2012 and more than 5,500 tweets were published every second as a result of tsunami and earthquake in Japan in 2011*) [8]. Manually checking each message and filtering those that are relevant is a heavy task, if not impossible. The link between social networks and disaster awareness has received increasing attention from researchers [2], [9], [10], [11]. These numerous studies have been conducted in recent years for the automatic identification of different types of social media content [9], [10], [11].

With the proliferation of social networks, an ongoing event is being discussed on all these channels. To get a complete view of the event, it is important to retrieve information from multiple sources. There are generally qualitative differences in the information obtained from different sources. However, when it comes to retrieving information posted on social networking sites and especially alerting those concerned for quick decision making, the task facing disaster managers is daunting. Several automated systems have been designed to help emergency managers identify and filter useful information posted on social media sites [12], [2].

The unambiguous nature of Twitter has allowed stakeholders to deliver relevant messages for the crisis and to access vast amounts of information that they may not have. However, the Web has several sources of information on which an ongoing event is discussed.

Several automated systems have been designed to help disaster managers identify and filter useful information posted on social networking sites [2]. Most of the work has focused on the use of social networks as a source of information on just a few phases of disaster management, including the response to the disaster, but few are dedicated to warning and all methods discuss information retrieval from social networks. Generally Twitter is the major source of information for these systems. The conception of systems for disaster management with different and various source of information particularly the social networking is a challenge.

In this article, we provide, not only, a solution to this challenge, but also, we propose better performances in term of recovery than other similar solutions.

The rest of paper is described as follows: section 2 presents the related work. In section 3, we describe our proposed model, we provide details on the imminence or disaster impact alert templates, information retrieval patterns from multiple sources, and preliminary results, followed a discussion of the results obtained. Finally, we conclude and give some future works.

II. Related Works

This section presents the most relevant related work for us, namely:

- Alert models
- Information retrieval models, in general, and those from several sources, in particular.

A. Alert models

De Albuquerque & al. [21] used only statistical analysis to identify general spatial patterns in the occurrence of flood-related tweets that may be associated with the proximity and severity of floods. While Robinson & al. [29] presented a notification system for identifying earthquakes, but also, in first-hand reports published on Twitter from tweets of target areas in Australia and New Zealand.

On the other hand, with the reinforcement of the use of microblogging platforms such as Twitter during the sudden onset of a natural or anthropogenic disaster, thousands of disaster-related messages posted. These online messages contain important information that can also be helpful in making quick decisions to help the affected community if they are dealt with quickly and effectively [1], [13]. Many types of processing techniques ranging from machine learning to natural language processing to computational linguistics have been developed [14] for different purposes [15], without, however, fully exploiting this data, despite the existence of some resources, such as annotated data and standardized lexical resources [16], [7], [3]. In [17], Terpstra & al. present a real-time analysis of Twitter data. But they use predefined geographic displays, the content of Twitter messages, and so on to analyze and filter disaster information.

Most event detection methods are based on keywords [18], such as Imran & al. [12], which analyze key words used in tweets during catastrophic events to classify messages as real-time event reports, using a support vector machine (SVM). This related research includes the methodology of Vieweg & al. [10] and disaster ontology to identify also tweets that provide only situational awareness. Kongthon & al. [19] develop classifiers to analyze only the tweets of the floods in Thailand in 2011. Also, Starbird & al. [4] and Vieweg & al. [5] analyzed the use of the microblog and the life cycle of disasters [12].

Bella Robinson & al. [20] developed a notification system alerting earthquakes, but, from, only, first-hand reports published on Twitter. João Porto de Albuquerque & al. [21] examined Twitter platform microblogging messages (tweets) produced during the flood of the Elbe in June 2013 in Germany, but, using only, statistical analysis to identify general spatial patterns in the occurrence of flood-related tweets that may be associated with the proximity and severity of floods.

Our work is most closely related to that of Koustav Rudra & al. [22], who studied Community Tweets, that is, abusive messages targeting specific religious / racial groups displayed during natural disasters. Except that they proposed methods to counter them by using anti-communal tweets published by some users during catastrophic events.

B. Models for retrieving information from multiple sources

Vulic & al. [11] use comparable document-aligned data to easily learn patterns of common topics or word concepts, in a bilingual recovery. However, the preparation of comparable documented data requires a lot of human involvement and time. Therefore, such approaches are not appropriate when information about an ongoing event must be retrieved quickly.

The work of Joseph A. Shaw and Edward A. Fox [23] focuses on methods for recovering separately from different sources and then combining them using data fusion techniques.

Multi-view recovery based on deep learning that attempts to learn document embedding in a common space, where differences between different data sources would not exist [24]. Unfortunately, this approach has low performance scores for all the methods used: indicating that the problem is difficult and requires better methods.

Also, according to Cynthia D. Balana [30], Google Crisis Response's (GCR) National Risk Management and Disaster Response Board (NDRRMC) had integrated Facebook and Twitter into its surveillance system and website for a quick response.

In this paper, we propose a new alert model for disaster management based on social networks. Our proposed model is based on <u>a neural network that integrates encapsulations from multiple sources, and thus retrieves information by combining multiple search results.</u>

II. New Alert Model to Disaster Management

Whether it's Twitter, Facebook or other social networks, these are platforms where people often express emotions. We detected emotions using four types of features: **1)** interjections, **2)** blasphemy, **3)** emoticons, and **4)** the general feeling of the message.

Interjections, blasphemies and emoticons are widely used by individuals to convey emotions such as anger, surprise, happiness, etc. To identify these three types of features, we used a combination of POS tags in the English tagger (*which contains tags for interjections, emoticons, etc.*), compiled lists of interjections and blasphemies on the Web for English, Spanish and French and regular expression patterns for emoticons [25].

Messages from Nepal's target regions are checked for frequency bursts of disaster keywords and processed to identify evidence of a disaster. The advantage of our Disaster Alert model is that it relies on evidence from Twitter's and Facebook's first-hand reports. This multi-view template associates an input document vector with a multi-view space, based on neural network, whose contextual information is available.

Let Ĥ be the non-empty set (Ĥ ≠ Ø) of the terms that we have compiled *(for example, terrorist act, explosion, earthquake, etc.*) that make up the set of keywords or hashtags for natural disasters and anthropogenic.

Let ei ∈ □ n, with i∈ [1, N], the incorporation of a content of the source message i relevant for, at least, a keyword or a hashtag hj ∈ Ĥ with j∈ [1, M]. □ n is the set of n contents.

We want to learn a generic space with the neural network

$$\check{E} = \{ek \text{ ; with } k \in [1, K] \} \qquad (1)$$

which normalizes the differences:

$$\check{E} = [E - \check{R} - \check{D}] \qquad (2)$$

where:

Ř is the set of retweets,

Ď is the set of duplicate contents,

and helps in better recovery of target content.

Thanks to the neural network, the transformation of {ei} into {ek} can be explained by:

{ei ==>ek = {ei \ ei is relevant for hj, ei ∉ Ř and ei ∉ Ď; and

hj ∈ Ĥ with j ∈ [1, M]}} (3)

where Ď and Ř represent respectively the set of duplicate retweets and contents.

The objective is then to maximize the size K of the set Ě.

In order to show the effectiveness of our model, we examined a specific event - the Nepal earthquake in April 2015 and the Queensland flood in 2013 - and post-event messages on two social media: Twitter and Facebook.

Tweets were collected using the Twitter Search API and Facebook posts were collected using the Radian6 tool.Both use the search keywords "nepal" and "earthquake" for the Nepal earthquake and the "Queensland" and "flood" keywords for the Queensland floods.

With the neural network and after de-duplication, we have:

a) 51,109 tweets,
b) and 76,338 messages on facebook

for the Nepal earthquake and

a) 36,077 tweets,
b) And 12,325 messages on Facebook

for the Queensland floods. All experiments reported here were performed on these two sets of data.

Standard relevance queries and judgments: we identified a set of disaster-specific information needs, as proposed in (F. Alam & al. (2018)) [26]. It's a set of keywords and hashtags for Facebook and Twitter respectively. Data collection and filtering are at the heart of disaster management using social media. The algorithms used to warn and alert depend on the messages published on the social media and their quality. Thus, every effort must be made to maximize the number of messages relevant to the disaster and to remove non-informative messages as proposed in (T. H. Nazer & al., (2017)) [31]. These contents are manually tagged (annotated) to remove those not related to the disaster (non-informative messages).

Table 1 presents the number of relevant documents found for each keyword or hashtag from the two datasets.

TABLE 1: EXAMPLES OF RELEVANT CONTENT FOR A SET OF KEYWORDS OR HASHTAGS.

Requests	Nepal earthquake		Queensland floods	
	Twitter	Facebook	Twitter	Facebook
Terrorist Acts	18	9	18	3
Explosion	155	50	255	157
Tsunamis	00	00	1543	111
Earthquake	1884	192	48	25

This content, obtained in this table, manually labeled (annotated) and cleaned from non-informative messages, will be used, in this space based neural network, to analyze live new messages through the learning allowed by this space based neural network, enriched by relevant messages and annotated manually.

The overall effectiveness of a large-scale disaster emergency response is a difficult entity to measure. Every disaster is different and every answer is different. In fact, similar events offer totally different results in their aftermath. For example, a category two hurricane can inflict much more damage than a category four, depending on the location of the impact. In addition, the response to an emergency in rural Mississippi will be totally different from the emergency response to a flood in New Orleans, Louisiana [27]. To effectively measure the overall response to these events, appropriate measures must be used and interpreted appropriately. Although there are many theories, it is very difficult to evaluate the effectiveness of a method because of the inherent irregularity of disasters. Two disasters are not alike and it is therefore difficult to determine if one method is better than another.

In a real disaster, the answer does not have to be a test of new ideas with inherent confusion, but rather a well-tested and realized answer [27]. The use of disaster datasets from social networks leads to ambiguous terms, inconsistent and incomplete data, and general confusion. Thus, the use of disaster data is fast and difficult at best when modeling and simulation are essential for an inexpensive and time-efficient method for observing a system and also for testing several inputs and different results [28]. One of the best reasons to implement a simulation for an observer is a system of "drawing the benefits of simulation in terms of cost and time". The importance of disaster simulation and emergency response was noted in [27]. The main objective of the experiment is the performance of the system under perilous circumstances. Calculations can be performed using, for example, Knowtator functionality [5], [10]. The annotations must be, then, carried out by voluntary annotators, of the same size and of the same class. Otherwise, the pilot alert system presented here consists of validating the research bases carried out as part of this work.

III. Conclusion & Perspectives

We present a new ad hoc real-time alert model to the approach or impact of a natural or anthropogenic disaster, based on a new multi-view recovery model from multiple sources. Such an approach is really useful for local disaster monitoring, but also for helping to make appropriate decisions.

The proposed model presented in this paper can completed with a series of new research questions and perspectives. (1) One simple way of future research is to use the alert model in civil or military situations. (2) An improvement can also come from the statistical validation of the information (*of the alert of the imminence or the impact of a possible disaster*) in order to avoid hoaxes. (3) The other improvement can also come from the model that will be based on a neural network that integrates encapsulations from multiple sources, and thus retrieves information by combining several search results. (4) This work can, also, be improved by generalizing the collection of data from, not just social networks, but also, from all social media. This will validate an alert for a possible imminent disaster, whether natural or anthropogenic.

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